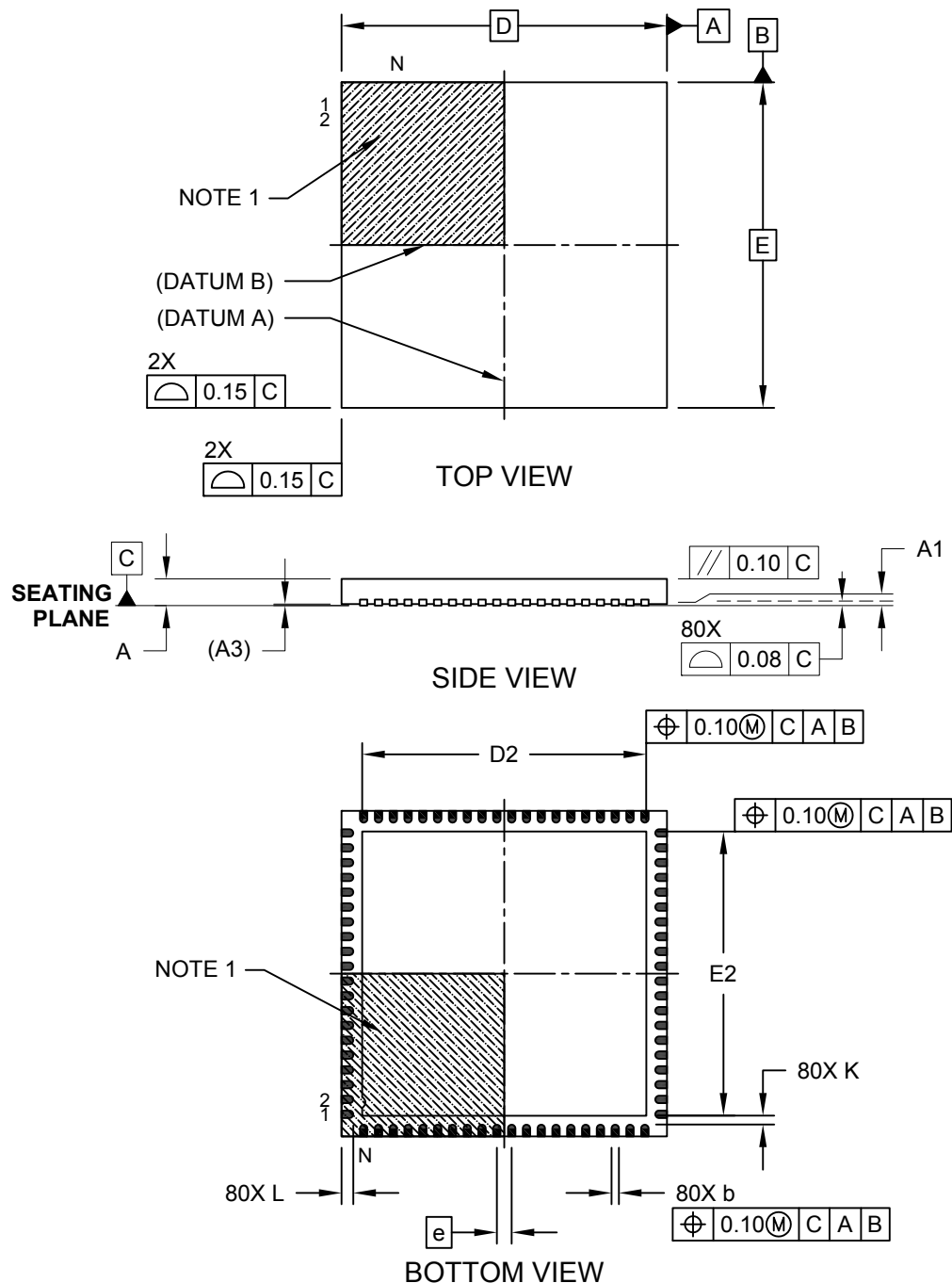


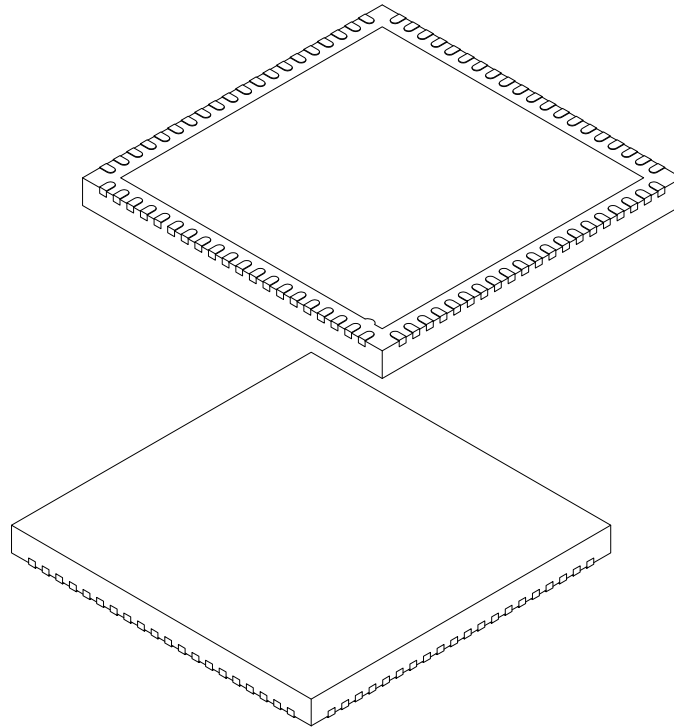
# 80-Lead Plastic Quad Flat, No Lead Package (6FX) – 11x11x0.9 mm Body [VQFN]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 80-Lead Plastic Quad Flat, No Lead Package (6FX) – 11x11x0.9 mm Body [VQFN]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



| Units                   |      | MILLIMETERS |      |      |
|-------------------------|------|-------------|------|------|
| Dimension Limits        |      | MIN         | NOM  | MAX  |
| Number of Terminals     | N    | 80          |      |      |
| Pitch                   | e    | 0.50 BSC    |      |      |
| Overall Height          | A    | 0.80        | 0.85 | 0.90 |
| Standoff                | A1   | 0.00        | -    | 0.05 |
| Terminal Thickness      | (A3) | 0.203 REF   |      |      |
| Overall Width           | E    | 11.00 BSC   |      |      |
| Exposed Pad Width       | E2   | 9.50        | 9.60 | 9.70 |
| Overall Length          | D    | 11.00 BSC   |      |      |
| Exposed Pad Length      | D2   | 9.50        | 9.60 | 9.70 |
| Terminal Width          | b    | 0.18        | 0.25 | 0.30 |
| Terminal Length         | L    | 0.30        | 0.40 | 0.50 |
| Terminal-to-Exposed Pad | K    | 0.20        | -    | -    |

### Notes:

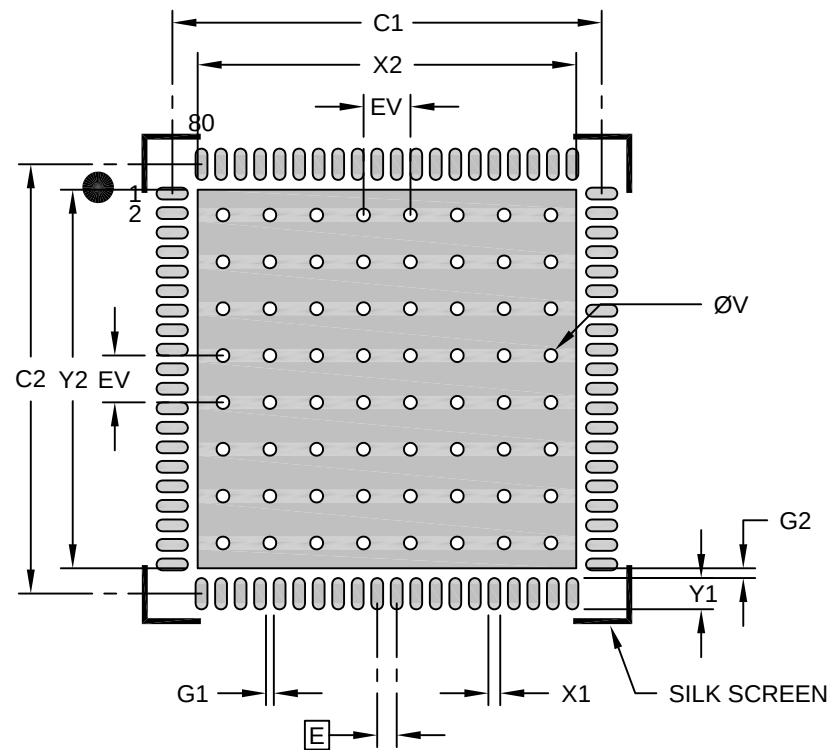
- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated.
- Dimensioning and tolerancing per ASME Y14.5M.

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

REF: Reference Dimension, usually without tolerance, for information purposes only.

## 80-Lead Plastic Quad Flat, No Lead Package (6FX) – 11x11x0.9 mm Body [VQFN]

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### RECOMMENDED LAND PATTERN

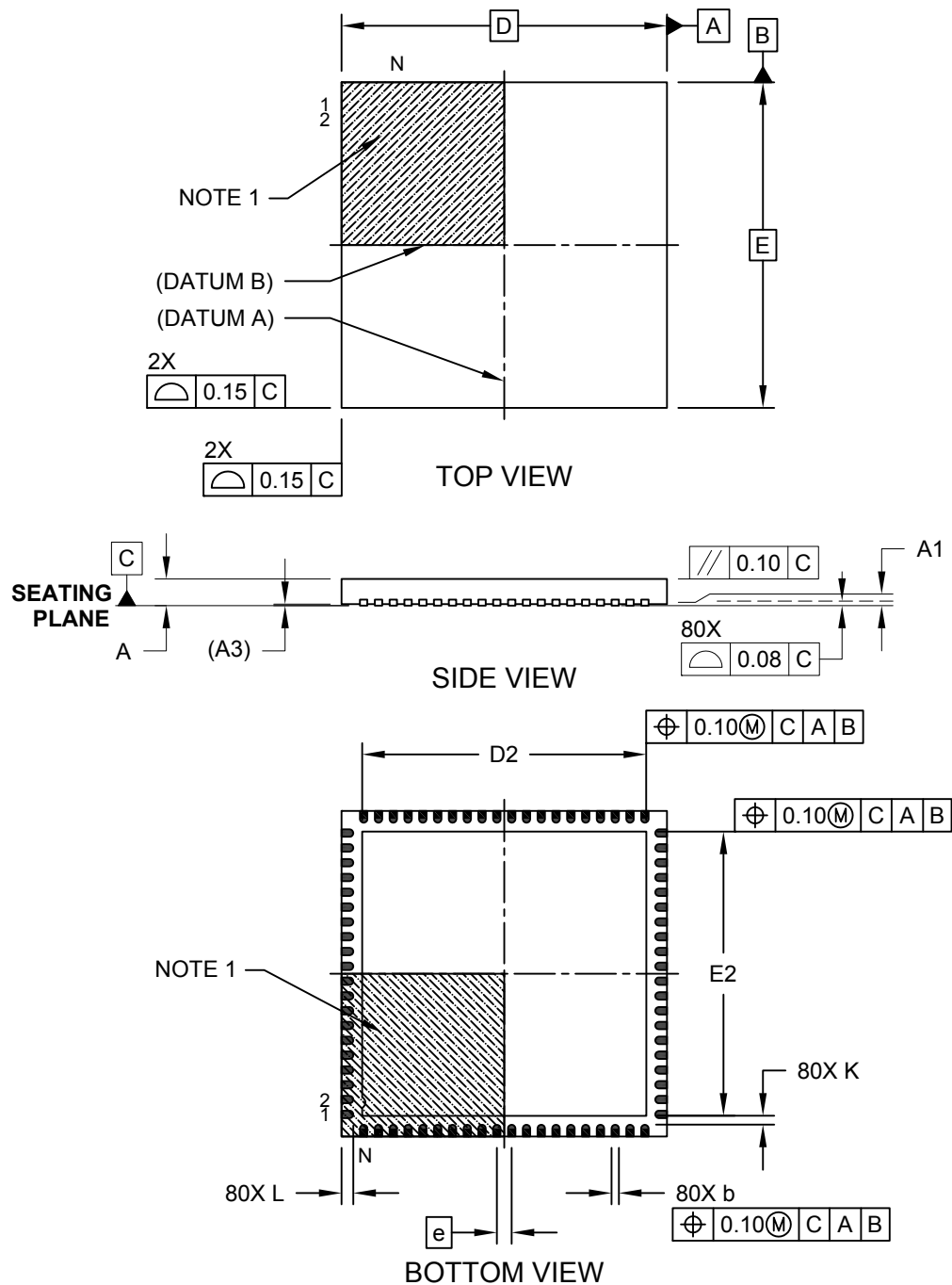
| Units                            |    | MILLIMETERS |       |      |
|----------------------------------|----|-------------|-------|------|
| Dimension Limits                 |    | MIN         | NOM   | MAX  |
| Contact Pitch                    | E  | 0.50 BSC    |       |      |
| Optional Center Pad Width        | X2 |             |       | 9.70 |
| Optional Center Pad Length       | Y2 |             |       | 9.70 |
| Contact Pad Spacing              | C1 |             | 11.00 |      |
| Contact Pad Spacing              | C2 |             | 11.00 |      |
| Contact Pad Width (X80)          | X1 |             |       | 0.30 |
| Contact Pad Length (X20)         | Y1 |             |       | 0.80 |
| Contact Pad to Contact Pad (X76) | G1 | 0.20        |       |      |
| Contact Pad to Center Pad (X80)  | G2 | 0.20        |       |      |
| Thermal Via Diameter             | V  |             | 0.33  |      |
| Thermal Via Pitch                | EV |             | 1.200 |      |

#### Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

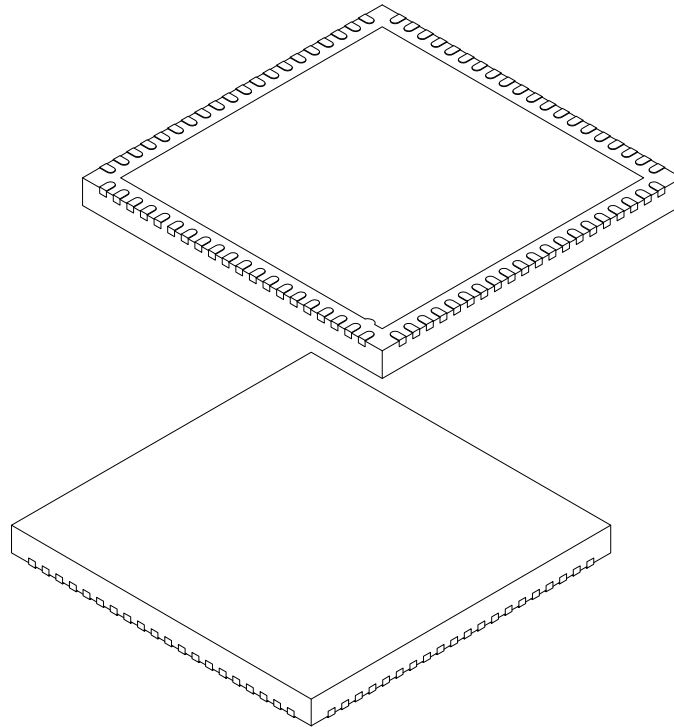
## 80-Lead Plastic Quad Flat, No Lead Package (8FX) – 11x11x0.9 mm Body [VQFN]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 80-Lead Plastic Quad Flat, No Lead Package (8FX) – 11x11x0.9 mm Body [VQFN]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



| Units                   |      | MILLIMETERS |      |      |
|-------------------------|------|-------------|------|------|
| Dimension Limits        |      | MIN         | NOM  | MAX  |
| Number of Terminals     | N    | 80          |      |      |
| Pitch                   | e    | 0.50 BSC    |      |      |
| Overall Height          | A    | 0.80        | 0.85 | 0.90 |
| Standoff                | A1   | 0.00        | -    | 0.05 |
| Terminal Thickness      | (A3) | 0.203 REF   |      |      |
| Overall Width           | E    | 11.00 BSC   |      |      |
| Exposed Pad Width       | E2   | 9.50        | 9.60 | 9.70 |
| Overall Length          | D    | 11.00 BSC   |      |      |
| Exposed Pad Length      | D2   | 9.50        | 9.60 | 9.70 |
| Terminal Width          | b    | 0.18        | 0.25 | 0.30 |
| Terminal Length         | L    | 0.30        | 0.40 | 0.50 |
| Terminal-to-Exposed Pad | K    | 0.20        | -    | -    |

### Notes:

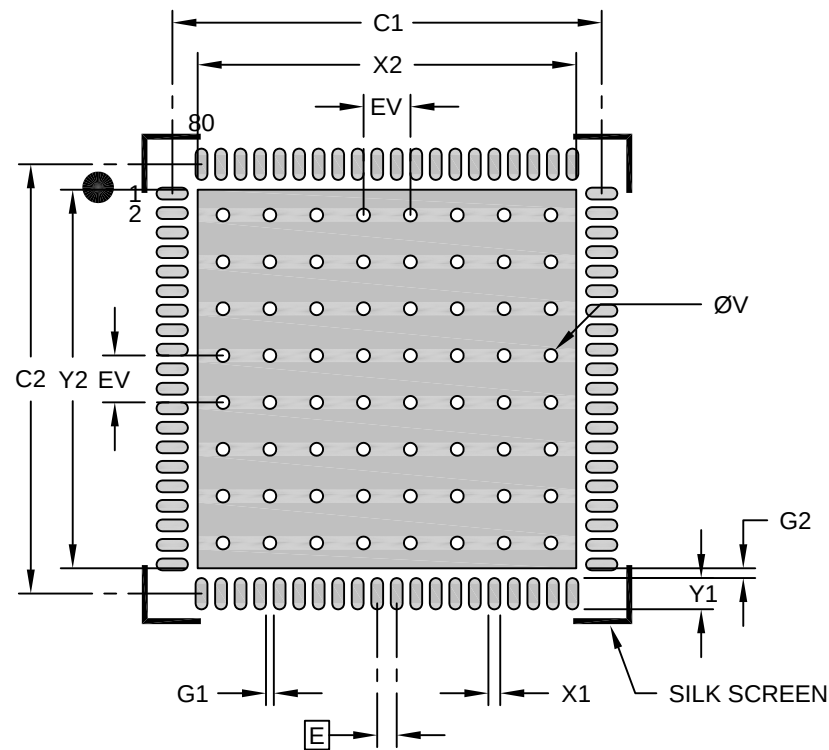
- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated.
- Dimensioning and tolerancing per ASME Y14.5M.

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

REF: Reference Dimension, usually without tolerance, for information purposes only.

## 80-Lead Plastic Quad Flat, No Lead Package (8FX) – 11x11x0.9 mm Body [VQFN]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Units                            |    | MILLIMETERS |       |      |
|----------------------------------|----|-------------|-------|------|
| Dimension Limits                 |    | MIN         | NOM   | MAX  |
| Contact Pitch                    | E  | 0.50 BSC    |       |      |
| Optional Center Pad Width        | X2 |             |       | 9.70 |
| Optional Center Pad Length       | Y2 |             |       | 9.70 |
| Contact Pad Spacing              | C1 |             | 11.00 |      |
| Contact Pad Spacing              | C2 |             | 11.00 |      |
| Contact Pad Width (X80)          | X1 |             |       | 0.30 |
| Contact Pad Length (X20)         | Y1 |             |       | 0.80 |
| Contact Pad to Contact Pad (X76) | G1 | 0.20        |       |      |
| Contact Pad to Center Pad (X80)  | G2 | 0.20        |       |      |
| Thermal Via Diameter             | V  |             | 0.33  |      |
| Thermal Via Pitch                | EV |             | 1.200 |      |

#### Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process